# Urban Electricity Demand Forecasting with a Hybrid Machine Learning Model

Ligang Hou
*Haimen Power Supply Branch, State Grid Jiangsu Electric Power Co, Ltd*
Haimen 226100, China
1223858626@qq.com

Xin Shen
*Haimen Power Supply Branch, State Grid Jiangsu Electric Power Co, Ltd*
Haimen 226100, China
2942060815@qq.com

Liqi Zhang
*Haimen Power Supply Branch, State Grid Jiangsu Electric Power Co, Ltd*
Haimen 226100, China
frank_zhlq@foxmail.com

2023 International Conference on Networking, Sensing and Control (ICNSC) | 979-8-3503-6950-2/23/$31.00 ©2023 IEEE | DOI: 10.1109/ICNSC58704.2023.10319027

***Abstract*—Urban electricity demand forecasting in the mid- and long-term with high accuracy is vital for the power systems' operation and planning. Many factors of uncertainties and non-linearities, in addition to economic trends and seasonal cycles, should be taken into account during forecasting. This study presents a hybrid machine learning model, i.e., ARIMA-LSTM (ARLS), for accurate mid- and long-term urban electricity demand forecasting. The model adopts the following three-fold ideas: a) incorporating various factors that affect electricity load, such as economic trends and seasonal cycles, and then selecting important features from them; b) employing an autoregressive integrated moving average (ARIMA) model to decompose the time series into level, trend, and seasonal components, and a long short-term memory (LSTM) network to account for the relevant non-linear features; and c) fusing the outputs of each learning module by using ensemble mechanism to enhance the forecast accuracy. By doing so, it combines the advantages of both traditional and ML approaches for time-series analysis. The model is tested on the monthly electricity consumption data of two Chinese cities, and the results indicate that it surpasses the existing state-of-the-art models in terms of performance.**

***Keywords—Urban Electricity Demand Forecasting, Hybrid Model, Machine Learning, Time Series Forecasting, Ensemble Learning***

## I. INTRODUCTION

An ideal smart grid is one that can achieve equilibrium between the electricity supply and demand in a target area. This can bring multiple benefits, such as optimizing the power distribution among different users and regions, generating economic benefits by reducing the cost of electricity production and transmission, and enhancing the social benefits by avoiding large-scale blackouts that may disrupt the normal functioning of society. These blackouts can be caused by various uncertainties [1], such as seasonal peaks that create a high demand for electricity, natural disasters that damage the power infrastructure, terrorist attacks that target the power grid, and urban unrest that affects the stability of the power system.

A key requirement for achieving smart grids is to forecast the urban electricity load accurately, especially for the mid- and long-term periods that are crucial for the efficient operation and planning of power systems. The mid- and long-term urban electricity demand (MLUED) forecasting is a complex and difficult task, as it involves a non-linear time series trend forecast that is influenced by many factors. Some of these factors are the economic development rate of a country, which reflects the electricity consumption patterns of different sectors, the seasonal weather cycles, which affect the heating and cooling demand of buildings, and various uncertainties that may arise from different sources [2], such as natural disasters, policy changes, or social events. Therefore, to enhance the accuracy of forecasting, it is important to identify and characterize the features that have a significant impact on the mid- and long-term urban electricity load.

In general, classical time series methods and emerging machine learning (ML) methods are the two main categories of existing methods for MLUED forecasting [3]. The models of time series methods, e.g., autoregressive integrated moving average (ARIMA) [4], linear regression [5], and exponential smoothing (ETS) [6], are relatively simple, efficient, and robust models that can deal with seasonal patterns well. Since they usually reflect strong domain knowledge, they possess high interpretability. But they have limited learning capacity and expressiveness, and fail to capture the long-term dependencies among snippets and thus cannot produce accurate MLUED forecasting at fine temporal resolutions.

ML methods like latent factor analysis models [7]-[10] and neural networks [11], [12] learn linear and non-linear features and mine complex data well, which makes an ML model a popular emerging technology for MLUED forecasting [13]-[15]. In particular, they can learn nonlinear representations from a large amount of data that records the historical load variations over time, thus enabling them to capture the complex patterns hidden in the raw time-series. For instance, an LSTM network is an ML model that learns temporal patterns from the load-series signals in a sequential way. It uses carefully designed gate mechanisms to balance the short-term and long-term memory of patterns [16]. However, ML models such as LSTM, are mainly driven by statistical learning theories or cognitive science, without incorporating any domain knowledge. Hence, their reasoning or inference processes are not interpretable, which limits their reliability and practicality in sensitive real applications like power systems.

Commonly, urban electricity demand is affected by many factors. To forecast the MLUED accurately, all the factors should be considered and their importance should be analyzed. Particularly, those features that matter for sure should be characterized accurately. To do this, this study introduces feature selection to filter out the important features affecting urban electricity demand. Inspired by the balance between traditional time-series methods and ML ones, it proposes a hybrid machine learning method called ARLS based on that for highly accurate MLUED forecasting. Firstly, to accurately predict urban electricity demand, it first selects the relevant features from the influencing factors. Secondly, it adopts ARIMA to extract the trend, seasonal, and level components of the electricity demand time series and linearly regresses them. It also adopts LSTM to regress the load-related features with a sequential, nonlinear neural network that focuses on uncertainties and non-linearities. Finally, it combines the acquired features of each learning module with an ensemble technique to achieve the final prediction results. By doing so, ARLS combines the advantages of both traditional and ML-based approaches for time-series analysis. It can incorporate domain knowledge easily and learn complex patterns from time-series data effectively, thus achieving highly accurate prediction results.

This study aims to make the following contributions:

a) A feature selection module. It filters out features that are irrelevant or redundant with redundancy analysis and correlation analysis, and reduces the data's dimensionality, which enhances the learning algorithms' performance.

b) A hybrid machine learning model, i.e., ARLS, for highly accurate MLUED forecasting. By combining the strengths of both conventional and machine learning methods for analyzing time-series data, ARLS can easily incorporate domain knowledge and effectively learn complex patterns from the data, resulting in highly accurate predictions.

Empirical studies on the monthly electricity consumption of two Chinese cities demonstrate that our model surpasses the existing state-of-the-art models in monthly electricity load forecast accuracy.

## II. PROBLEM STATEMENT

### A. Electricity Demand Time Series

The patterns of electricity demand change a lot over time, and they show different features, such as repeating cycles at different time scales, and directions of change. For instance, the electricity demand may vary according to the time of the day, the day of the week, the month of the year and the season of the year, and so on. Some patterns may occur only on a specific day of the week, while others may occur every day, such as the higher electricity demand in the morning due to more people being active. The presence of such features, especially when they are mixed with directions of change, makes the prediction of such time series very hard. Among other things, a good prediction model must capture features that have different time lengths.

### B. Problem Formulation

We can use the following Eq. (1) to model the MLUED forecasting based on time series analysis, which is considered as a generic regression model [17], i.e.,

$$\hat{y}_{t+1} = F\left(x_{t-T+1}, \ldots, x_t\right), \quad (1)$$

where $\{x_{t-T+1}, \ldots, x_t\}$ denotes the feature of $d$-dimensional time series in $i$-th moving window, which includes $T$ time steps before time $(t+1)$; and $\hat{y}_{t+1}$ denotes the prediction of time $(t+1)$'s output values.

We want to make the prediction values $\hat{y}$ as close as possible to the true values $y$, which we get from historical data in training and measurements in testing. This means we need to minimize the distance between them, as formulated in (2)

$$\min_F \sum_t Distance\left(y_t, \hat{y}_t\right), \quad (2)$$

where $t$ can be considered as any time period, and $Distance$ can be considered as any distance measure.

## III. PROPOSED FORECASTING MODEL

By incorporating feature engineering, and combining the acquired features of ARIMA and LSTM, the ARLS model creates a hybrid machine learning method highly accurate MLUED forecasting, which is described as follows.

### A. Overall Framework

An overall framework of the proposed ARLS model is depicted in Fig. 1.

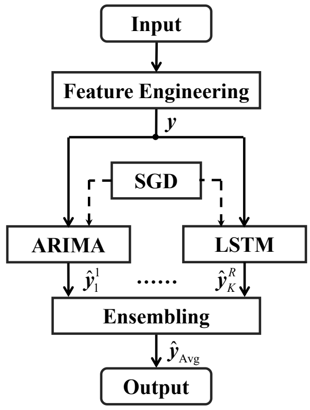


Fig. 1. Overall framework of the proposed ARLS model.

As shown in Fig. 1, the ARLS model consists of five main components:

- Feature Engineering. Feature selection is achieved by performing relevance and redundancy analyses on each component to select the most relevant features for each component. In detail, it receives various features related to the urban electricity load and chooses the most relevant features for the next modeling.
- Time Series Module (ARIMA). The trend component demonstrates a significant long-term rising trend over time and reveals nonlinear features, and the ARIMA is employed to fit the long-term rising trend to ensure generalization ability.
- ML Module (LSTM). An LSTM network learns from the information at each timestamp and preserves it for future use. This helps the network to make better predictions of time series events. Besides, it has good approximation capabilities. Hence, LSTM is adopted to regress the load-related features.
- Ensembling. By combining the acquired features of each module via an ensemble technique to achieve the final prediction results, ARLS is able to combine the advantages of both classical and ML-based approaches for time-series analysis, thus achieving highly accurate prediction results.
- Optimization. The Stochastic gradient descent (SGD) algorithm is adopted to optimize the developed ARLS model for minimizing the prediction error.

### B. Feature Engineering

The trend component shows a steady increase and the seasonal component changes periodically, influenced by factors like the economy and the weather. The irregular component, on the other hand, captures the nonlinear noise that represents random factors such as abrupt changes in temperature.

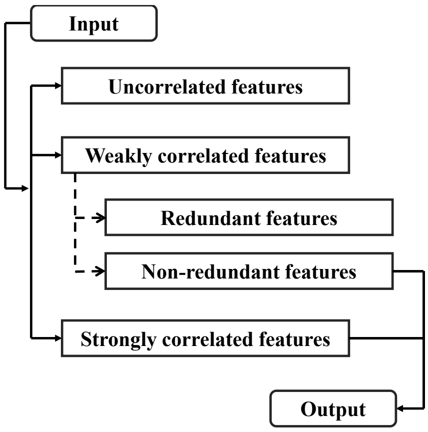


Fig. 2. Overall flowchart of feature engineering.

Many relevant features are fed into the module, such as holidays, weather, humidity, wind, rainfall, barometric pressure, cloudiness, maximum and minimum temperature, etc. The feature selection [18] is an important module for ARLS that lowers the dimensionality of the dataset and eliminates uncorrelated and redundant features by leveraging relevance and redundancy analyses to choose the most relevant features for each component. Its overall flowchart is depicted in Fig. 2. Specifically, the relationship between the variables and the targets is tested by relevance analysis using a statistical test. The linear correlation between the features and the trend or seasonal components is assessed by the Pearson Correlation Coefficient (PCC), whose value goes in the range of [−1, 1], and it shows no relationship when it is zero. The formula for computing it is:

$$PCC = \frac{\sum_i \left(X_i - \bar{X}\right)\left(Y_i - \bar{Y}\right)}{\sqrt{\sum_i \left(X_i - \bar{X}\right)^2}\sqrt{\sum_i \left(Y_i - \bar{Y}\right)^2}}, \tag{3}$$

where a feature and the targeted component are represented by $X_i$ and $Y_i$ for the $i$-th sample; and the mean values of $n$ samples of $X_i$ and $Y_i$ are represented by $\bar{X}$ and $\bar{Y}$, respectively. Copula entropy is employed to measures the nonlinear correlations between features and the irregular component. The marginal distribution conditions of $X_i$ and $Y_i$ are represented by $H(X_i)$ and $H(Y_i)$, and the Copula entropy can be computed as

$$E_c\left(X_i, Y_i\right) = -\iint c\left(H\left(X_i\right), H\left(Y_i\right)\right) dH\left(X_i\right) dH\left(Y_i\right), \tag{4}$$

where $c(H(X_i), H(Y_i))$ denotes the probability density function of Copula function, which is calculated as

$$c\left(H\left(X_i\right), H\left(Y_i\right)\right) = \frac{dC\left(H\left(X_i\right), H\left(Y_i\right)\right)}{dH\left(X_i\right) dH\left(Y_i\right)}. \tag{5}$$

By combining Eqs. (16)-(18), the relevant features can be selected.

## C. ARIMA Model

A traditional time-series forecasting model is ARIMA (autoregressive integrated moving average) [19]. It consists of autoregressive, moving average, and autoregressive moving average models. It works by first determining the stationary degree, then stabilizing the non-stationary time-series with the difference method, and finally choosing AR($p$) and MA($q$) to decide the order of the model. The general formulation of the ARIMA($p, d, q$) is given by

$$\begin{aligned} w_t = {} & \phi_1 w_{t-1} + \phi_2 w_{t-2} + \cdots \phi_p w_{t-p} \\ & + \phi + u_t - \theta_1 u_{t-1} - \theta_2 u_{2-1} - \cdots \theta_q u_{t-q}. \end{aligned} \tag{6}$$

where ($\phi_1$, $\phi_2$, …, $\phi_p$) is a set of autoregressive (AR) coefficients and $p$ represents the order of AR; ($\theta_1$, $\theta_2$, …, $\theta_q$) is a set of moving average (MA) coefficients and $p$ represents the order of MA; $u_t$ denotes a white noise sequence which can be zero mean, normal distribution, and independent from each other. Note that the values of $p$ and $q$ can be decided by using autocorrelation and bias correlation plots, $d$ is the order for which the data needs to be differenced.

## D. LSTM Network

As a type of RNN, LSTM adopts a "feedback" connection to pass the previous step's output as the current step's input, creating an artificial memory. It solves the issue of vanishing gradient via replacing RNN nodes with memory cells and gating mechanism. Three gates, i.e., the input gate $\Gamma_i$, the forget gate $\Gamma_f$ which is adopted to update the cell memory $c_t$, and the output gate $\Gamma_o$ which is adopted to update the state cell $h_{t-1}$, control the cell state in LSTM memory cell based on the input $x_t$ and the previous output $h_{t-1}$. They can adjust the cell state by remembering, forgetting or scaling it. The main principle of LSTM is formulated as

$$\begin{cases} \tilde{c}_t = \tanh\left(W_c\left(h_{t-1}, x_t\right) + b_c\right); \\ I_t = \sigma\left(W_i\left(h_{t-1}, x_t\right) + b_i\right); \\ F_t = \sigma\left(W_f\left(h_{t-1}, x_t\right) + b_f\right); \\ O_t = \sigma\left(W_o\left(h_{t-1}, x_t\right) + b_o\right); \\ c_t = I_t \cdot \tilde{c}_t + F_t \cdot c_{t-1}; \\ h_t = O_t \cdot \tanh\left(c_t\right); \\ \hat{y}_t = \sigma\left(W_p \cdot h_t\right) + b_p, \end{cases} \tag{7}$$

where $W_c$, $W_i$, $W_f$, $W_o$, $W_p$ denote the weight matrices, and $b_c$, $b_i$, $b_f$, $b_o$, $b_p$ denote the bias vectors. Note that tanh(·) and $\sigma$(·) are activation functions, and $\tilde{c}_t$ is an intermediate cell state.

## E. Ensembling

Ensembling [20] is a common technique to enhance weak learners. They combine multiple learning algorithms in some way to produce a shared output, which can offer better accuracy and stability than individual learners. As discussed in Section III-A, an ARIMA model is firstly employed to fit the long-term rising trend of MLUED time series to ensure the generalization ability. It has limited learning capacity and expressiveness, and fails to capture long-term dependencies among snippets. An LSTM network uses carefully designed gate mechanisms to balance the short-term and long-term memory of patterns. However, it is mainly driven by statistical learning theories or cognitive science, without incorporating any domain knowledge. To take the best of the advantages of traditional and ML-based approaches for time-series analysis, we aim to combine ARIMA and LSTM to build an MLUED forecasting model that can possess the merits of both domain knowledge and nonlinear representation learning ability.

TABLE I. THE STATISTICAL INFORMATION OF THE TESTED DATASETS

| No. | Time range | Features |
|---|---|---|
| D1 | 01/01/2013-31/12/2021 | 12 |
| D2 | 01/01/2013-31/12/2021 | 12 |

In this study, the proposed ARLS model adopts multiple sources of discreteness, i.e., a) the random training process via SGD; b) the random sampling of the training set of involved datasets to train each model; and c) different hypotheses to initialize the decision parameters. Hence, the final results of ARLS can be achieved by integrating at three levels:

a) Training level: It averages the results outputted in the most recent *L* iteration epochs.

b) Data level: It averages the results obtained by the pool of *K* forecasting models built on subsets of the training set.

c) Feature level: It averages the features from the adopted ARIMA and LSTM to generate the final prediction values.

## IV. EXPERIMENTAL RESULTS

### A. General Settings

#### 1) Datasets

This study takes the monthly electricity consumption data from two Chinese cities as benchmark datasets, numbered as D1 and D2, to evaluate the ARLS's performance on MLUED forecasting. The datasets include 12 features related to time series, such as holidays, weather conditions, humidity, wind speed, rainfall, air pressure, cloud cover, and maximum and minimum temperatures. Table I records the statistics of the involved datasets. Note that the last 12 months of data, from Jan. to Dec. 2021, are used as the test set in this study, while the remaining data is used for training.

#### 2) Evaluation Metrics

Considering the evaluation of the proposed ARLS model's performance on MLUED forecasting, one major concern is how close the achieved prediction values are to the ground truth values. To this end, we evaluate the prediction accuracy of all tested methods with two widely-used metrics [21]-[23], i.e., mean absolute error (MAE) and mean absolute percentage error (MAPE). Note that MAE reflect the absolute deviation between prediction values and the corresponding true values; and MAPE reflects the average ratio of prediction error. Their calculations are given by

$$\mathrm{MAE}=\frac{1}{N}\sum_{i=1}^{N}\left|y_i-\hat{y}_i\right|,\mathrm{MAPE}=\frac{1}{N}\sum_{i=1}^{N}\left|\frac{y_i-\hat{y}_i}{\hat{y}_i}\right|\times 100\%. \qquad (8)$$

In Eq. (8), *N* denotes the number of data points in the time series, and $|\cdot|$ calculates the absolute value.

#### 3) Tested Models

In our experiments, the proposed ARLS model is assessed by comparing with seven relevant state-of-the-art methods. Note that those comparison models can be divided into three categories, including a time series analysis model, i.e., ETS [6]; four ML-based models, i.e., GRNN [15], MLP [24], XGBoost [25], and SVR [26]; and two hybrid models, i.e., NBEATS [2] and APLF [27]. All of them possess different characteristics, and their descriptions are briefly summarized in Table II, and the corresponding hyperparameter settings are recorded in Table III.

TABLE II. THE DESCRIPTIONS OF THE TESTED MODELS

| Model | Description |
|---|---|
| ETS | It performs multiplicative seasonal decomposition on a time series, extracting both level and seasonal components. |
| NBEATS | It is a deep learning architecture that uses fully connected layers and forward and backward residual connections. |
| MLP | It has a hidden layer and Sigmoid neurons. |
| GRNN | It is a four-layer neural network with Gaussian nodes focused on the training model. |
| SVR | It is a tolerance regression model with strict linear regression. |
| XGBoost | It is a library for distributed gradient enhancement that has been optimized for efficiency, flexibility, and portability. |
| APLF | It is a method for forecasting probabilities that uses adaptive online learning of hidden Markov models. |

TABLE III. HYPERMETER SETTINGS OF THE TESTED MODELS

| Model | Description |
|---|---|
| ETS | $\alpha$=0.46, $\beta$=0, $\gamma$=0. |
| NBEATS | Batchsize=10, learning_rate=0.001, epoch=100, layers=3, blocks=3, lookback period=12, optimizer=Adam. |
| MLP | Learning rate=0.001, max_iter=50, $\alpha$=5e-2. |
| GRNN | Spread=10. |
| SVR | Kernel='linear', epsilon=0.3, *C*=1. |
| XGBoost | Learning rate=0.02, max_depth=3, estimator number=1000. |
| APLF | $\lambda_d$=0.2, $\lambda r$=0.5, *C*=12, *R*=4. |
| ARLS | *p*=3, *q*=1, *d*=1; batchsize=3, learning_rate=0.01, epoch=30, period=6 ,optimizer=Adam. |

### B. Comparison Results

#### 1) Comparison Results on MAE

In terms of MAE, comparison results among the proposed ARLS model and seven compared models on D1 and D2 are depicted in Figs. 3 and 4, respectively. According to them, we achieve the following important findings:

a) In general, the prediction errors in months 10-12 are lower than those in months 1-3 and 7-8. The results indicate that the urban electricity loads from October to December are relatively stable.

b) Both traditional time series analysis models such as ETS and the ML-based one such as XGBoost achieve very high prediction errors, while the hybrid learning model such as NBEATS obtains the lower prediction error than them. Such results tell us that neither traditional time series analysis nor

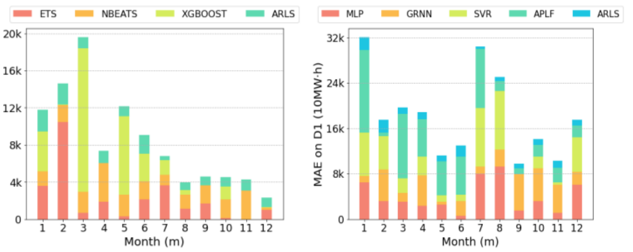


Fig. 3. Comparison results regarding MAE on D1.

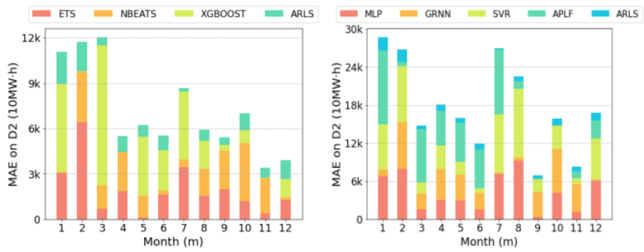


Fig. 4. Comparison results regarding MAE on D2.

TABLE IV. COMPARISON RESULTS OF MAPE ON D1

| Month | ETS | NBEATS | MLP | GRNN | SVR | XGBOOST | APLF | ARLS |
|---|---|---|---|---|---|---|---|---|
| 1 | 7.05% | 3.08%● | 12.69% | 2.23%● | 14.84% | 8.40% | 28.46% | 4.48% |
| 2 | 35.19% | 6.15%● | 10.67% | 18.81% | 19.53% | 0.11%● | 2.30%● | 7.55% |
| 3 | 1.43%● | 4.76% | 6.41% | 3.38% | 5.39% | 32.80% | 24.04% | 2.45% |
| 4 | 4.37% | 9.64% | 5.48% | 12.40% | 7.57% | 0.01%● | 15.21% | 2.99% |
| 5 | 0.63%● | 5.37% | 5.93% | 1.21%● | 2.53% | 19.42% | 13.71% | 2.44% |
| 6 | 4.69% | 4.22%● | 1.39%● | 5.60% | 2.40%● | 6.45% | 14.55% | 4.40% |
| 7 | 6.59% | 2.14% | 14.69% | 2.26% | 18.81% | 2.89% | 19.04% | 0.75% |
| 8 | 2.07% | 2.69% | 16.66% | 5.28% | 18.70% | 0.91%● | 3.06% | 1.40% |
| 9 | 3.73% | 4.47% | 3.48% | 14.28% | 0.05%● | 0.01%● | 2.13% | 1.99% |
| 10 | 0.34%● | 4.66% | 7.50% | 13.75% | 5.01% | 3.30% | 4.88% | 2.49% |
| 11 | 0.04%● | 7.06% | 2.51%● | 11.41% | 0.96%● | 0.00%● | 5.96% | 2.65% |
| 12 | 2.05% | 0.40%● | 12.07% | 4.65% | 12.09% | 0.21%● | 4.25% | 1.96% |
| Avg-MAPE | 5.68%±2.24% | 4.55%±2.42% | 8.29%±4.97% | 7.94%±5.86% | 8.99%±7.42% | 6.21%±10.1% | 11.47%±9.05% | 2.96%±2.11% |
| Win/Loss | 4/8 | 4/8 | 2/10 | 2/10 | 3/9 | 6/6 | 1/11 | 62/22 |
| F-rank | 3.583 | 4.083 | 5.333 | 5.333 | 5.333 | 3.417 | 6.083 | 2.833 |
| *p*-value | 0.2349 | **0.0386** | **0.0024** | **0.0061** | **0.0081** | 0.3386 | **0.0034** | - |

TABLE V. COMPARISON RESULTS OF MAPE ON D2

| Month | ETS | NBEATS | MLP | GRNN | SVR | XGBOOST | APLF | ARLS |
|---|---|---|---|---|---|---|---|---|
| 1 | 6.84% | 0.18%● | 15.23% | 2.23%● | 15.99% | 13.01% | 26.01% | 4.79% |
| 2 | 21.20% | 11.02% | 26.13% | 24.52% | 29.13% | 0.21%● | 2.26%● | 6.34% |
| 3 | 1.68% | 3.85% | 3.91% | 6.08% | 4.29% | 22.95% | 20.91% | 1.36% |
| 4 | 5.06% | 7.14% | 8.22% | 13.58% | 10.04% | 0.01%● | 14.87% | 2.85% |
| 5 | 0.25%● | 3.85% | 7.86% | 10.90% | 5.45% | 10.59% | 16.53% | 2.01% |
| 6 | 4.10% | 0.71%● | 3.83% | 6.49% | 1.98%● | 6.75% | 15.34% | 2.43% |
| 7 | 7.11% | 1.06% | 14.54% | 0.72% | 18.73% | 9.22% | 21.10% | 0.50% |
| 8 | 3.12% | 3.59% | 18.50% | 1.17%● | 21.66% | 3.74% | 2.45% | 1.47% |
| 9 | 4.88% | 6.29% | 0.96%● | 9.63% | 4.93% | 0.99%● | 0.41%● | 1.22% |
| 10 | 3.35% | 10.95% | 11.72% | 19.62% | 10.51% | 2.49%● | 0.14%● | 3.20% |
| 11 | 1.01%● | 6.48% | 3.29% | 11.88% | 2.64% | 0.02%● | 3.06% | 1.82% |
| 12 | 2.91% | 0.40%● | 14.18% | 0.24%● | 15.05% | 2.90% | 6.62% | 2.87% |
| Avg-MAPE | 5.13%±2.7% | 4.63%±3.83% | 10.70%±7.41% | 8.92%±7.71% | 11.70%±8.54% | 6.07%±6.91% | 10.81%±9.30% | 2.57%±2.23% |
| Win/Loss | 2/10 | 3/9 | 1/11 | 3/9 | 1/11 | 5/7 | 3/9 | 66/18 |
| F-rank | 3.667 | 3.833 | 5.500 | 5.167 | 5.833 | 4.083 | 5.417 | 2.500 |
| *p*-value | **0.0134** | **0.032** | **0.0005** | **0.0081** | **0.0005** | 0.1167 | **0.0134** | - |

ML models can achieve good prediction accuracy alone, and a hybrid one can incorporate domain knowledge easily and learn complex patterns from time-series data, thus achieving accurate prediction results.

c) ARLS outperforms its peers for MLUED forecasting. From Figs. 3 and 4, we can see that our model always achieves the lowest prediction errors on most testing cases. The results validate that ensembling ARIMA and LSTM forms a complementary effect, so that the proposed ARLS model can obtain better prediction accuracy.

### 2) *Comparison Results on MAPE*

In this part, we aim to compare the models' performance with MAPE metric, and the comparison results on D1 and D2 are summarized in Tables IV and V, respectively. In addition, we also perform some statistical analysis on these comparison results, which is helpful in understanding them. Specifically, we summarize the average MAPE values over 12 months at the fourth-last row of each table. We summarize the win/loss scores at the third-last row of each table. To compare the accuracy of different models on various datasets, we used the Friedman test, which assigns smaller F-rank values to better models. The second-last row of each table shows the results of this test. Finally, we perform the Wilcoxon signed-rank test to see if the proposed model has a much lower MAPE than each of the other models. The last row of each table shows the significance level of this test, and values below the critical level of 0.05 are in bold. From these results in Tables IV and V, we obtain the following important findings:

a) ARLS outperforms the other models in most testing cases. For instance, ARLS achieves the lowest mean MAPE value of 2.96% on D1 as shown in Table IV, and that of 2.57% on D2 as shown in Table V. Besides, ARLS has fewer and more consistent errors than the other models, as it wins 62 out of 84 testing cases on D1 and 66 out of 84 testing cases on D2.

b) ARLS obtains the best accuracy on both datasets, as it achieves the smallest F-rank values among all the models.

c) ARLS has significantly higher prediction accuracy than each of the comparison models on both datasets, with only three testing cases having *p*-values greater than the critical level of 0.05. Despite the null hypotheses being rejected in three testing cases, our ARLS model still outperforms the other comparison models in terms of MAPE metric.

## V. CONCLUSION

A hybrid machine learning model ARLS is proposed for MLUED forecasting. It has four key steps: a) selecting the most important features from the factors that affect the urban electricity load; b) adopting ARIMA to capture the seasonal, trend and level components of the electricity load time series; c) adopting LSTM to mine relevant features and compensate for the drawbacks of time series methods; and d) ensembling both the time series analysis model, i.e., ARIMA, and the ML-based model, i.e., LSTM, to achieve effective aggregation of the performance of each module. Experimental results on the data of monthly electricity consumption of two cities in China in comparison with seven state-of-the-art MLUED forecasting models indicate that our method has higher forecast accuracy than its peers. To handle the missing data issue of the feature selection and improve the ability for the MLUED forecasting task, our future work plans to introduce some intelligent optimization algorithms such as evolution techniques [28]-[30] and latent factor analysis [31]-[38] into the model.


## VI. ACKNOWLEDGMENT

This paper is supported by the science and technology project of Jiangsu Electric Power Co, Ltd under the grant J2022007 (Artificial intelligence load forecasting technology considering high permeability of distributed new energy and meteorological comprehensive evaluation indicators. *Corresponding Author: Liqi Zhang*.


## REFERENCES


[1] T. Li, Y. Wang, and N. Zhang, "Combining probability density forecasts for power electrical loads," *IEEE Trans. on Smart Grid*, vol. 11, no. 2, pp. 1679-1690, March 2020.

[2] B. Oreshkin, et al., "N-BEATS neural network for mid-term electricity load forecasting," *Applied Energy*, vol. 293, no. 1, pp. 116918, 2021.

[3] A. Mamun, et al., "A comprehensive review of the load forecasting techniques using single and hybrid predictive models," *IEEE Access*, vol. 8, pp. 134911-134939, 2020.

[4] K. Sekiguchi, et al., "Autoregressive moving average jointly-diagonalizable spatial covariance analysis for joint source separation and dereverberation," *IEEE/ACM Trans. on Audio, Speech, and Language Processing*, vol. 30, pp. 2368-2382, 2022.

[5] D. Yuan, A. Proutiere and G. Shi, "Distributed online linear regressions," *IEEE Trans. on Information Theory*, vol. 67, no. 1, pp. 616-639, Jan. 2021.

[6] G. Xie, N. Zhang, and S. Wang, "Data characteristic analysis and model selection for container throughput forecasting within a decomposition-ensemble methodology," *Transportation Research Part E: Logistics and Transportation Review*, vol 108, pp.160-178, 2017.

[7] D. Wu and X. Luo, "Robust latent factor analysis for precise representation of high-dimensional and sparse data," *IEEE/CAA Journal of Automatica Sinica*, vol. 8, no. 4, pp. 796-805, April 2021.

[8] Z. Liu, X. Luo and Z. Wang, "Convergence analysis of single latent factor-dependent, nonnegative, and multiplicative update-based nonnegative latent factor models," *IEEE Trans. on Neural Networks and Learning Systems*, vol. 32, no. 4, pp. 1737-1749, April 2021.

[9] X. Luo, Y. Zhou, Z. Liu and M. Zhou, "Fast and accurate non-negative latent factor analysis of high-dimensional and sparse matrices in recommender systems," *IEEE Trans. on Knowledge and Data Engineering*, vol. 35, no. 4, pp. 3897-3911, 1 April 2023.

[10] X. Luo, H. Wu and Z. Li, "NeuLFT: A novel approach to nonlinear canonical polyadic decomposition on high-dimensional incomplete tensors," *IEEE Trans. on Knowledge and Data Engineering*, vol. 35, no. 6, pp. 6148-6166, 1 June 2023.

[11] S. Li, M. Zhou and X. Luo, "Modified primal-dual neural networks for motion control of redundant manipulators with dynamic rejection of harmonic noises," *IEEE Trans. on Neural Networks and Learning Systems*, vol. 29, no. 10, pp. 4791-4801, Oct. 2018.

[12] M. Liu, et al., "Activated gradients for deep neural networks," *IEEE Trans. on Neural Networks and Learning Systems*, vol. 34, no. 4, pp. 2156-2168, April 2023.

[13] H. Wen, et al., "Deep learning based multistep solar forecasting for pv ramp-rate control using sky images," *IEEE Trans. on Industrial Informatics*, vol. 17, no. 2, pp. 1397-1406, 2021.

[14] D. Cao, et al., "Robust deep Gaussian process-based probabilistic electrical load forecasting against anomalous events," *IEEE Trans. on Industrial Informatics*, vol. 18, no. 2, pp. 1142-1153, 2022.

[15] H. Bendu, B. Deepak, and S. Murugan, "Multi-objective optimization of ethanol fuelled HCCI engine performance using hybrid GRNN-PSO," *Applied Energy*, vol. 187, pp. 601-611, 2017.

[16] M. Fan, et al., "Short-term load forecasting for distribution network using decomposition with ensemble prediction," in *Proc. of the 2019 Chinese Automation Congress (CAC)*, 2019, pp. 152-157.

[17] G. Zhang and J. Guo, "A novel method for hourly electricity demand forecasting," *IEEE Trans. on Power Systems*, vol. 35, no. 2, pp. 1351-1363, March 2020.

[18] D. Wu, et al., "A latent factor analysis-based approach to online sparse streaming feature selection," *IEEE Trans. on Systems, Man, and Cybernetics: Systems*, vol. 52, no. 11, pp. 6744-6758, Nov. 2022.

[19] K. Yunus, T. Thiringer and P. Chen, "ARIMA-based frequency-decomposed modeling of wind speed time series," *IEEE Trans. on Power Systems*, vol. 31, no. 4, pp. 2546-2556, 2016.

[20] C. Lin, et al., "Ensemble method with heterogeneous models for battery state-of-health estimation," *IEEE Trans. on Industrial Informatics*, doi: 10.1109/TII.2023.3240920.

[21] F. Teng and G. Strbac, "Full stochastic scheduling for low-carbon electricity systems," *IEEE Trans. on Automation Science and Engineering*, vol. 14, no. 2, pp. 461-470, Apr. 2017.

[22] D. Wu, et al., "A data-characteristic-aware latent factor model for web services QoS prediction," *IEEE Trans. on Knowledge and Data Engineering*, vol. 34, no. 6, pp. 2525-2538, 1 June 2022.

[23] L. Hu, et al., "A distributed framework for large-scale protein-protein interaction data analysis and prediction using MapReduce," *IEEE/CAA Journal of Automatica Sinica*, vol. 9, no. 1, pp. 160-172, Jan. 2022.

[24] P. Pełka and G. Dudek, "Pattern-based forecasting monthly electricity demand using multilayer perceptron," *in Proc. of the 2019 Int. Conf. on Artificial Intelligence and Soft Computing*, 2019, pp. 663-672.

[25] J. Xie, et al., "A novel bearing fault classification method based on XGBoost: the fusion of deep learning-based features and empirical features," *IEEE Trans. on Instrumentation and Measurement.,* vol. 70, pp. 1-9, 2021

[26] Y. Chen et al., "Short-term electrical load forecasting using the Support Vector Regression(SVR) model to calculate the demand response baseline for office buildings," *Applied Energy*, vol. 195, pp. 659-670, 2017.

[27] V. Álvarez, S. Mazuelas, and J. A. Lozano, "Probabilistic load forecasting based on adaptive online learning," *IEEE Trans. on Power System.,* vol. 36, no. 4, pp. 3668-3680, 2021.

[28] L. Jin, L. Wei and S. Li, "Gradient-based differential neural-solution to time-dependent nonlinear optimization," *IEEE Trans. on Automatic Control*, vol. 68, no. 1, pp. 620-627, Jan. 2023.

[29] J. Chen, X. Luo and M. Zhou, "Hierarchical particle swarm optimization-incorporated latent factor analysis for large-scale incomplete matrices," *IEEE Trans. on Big Data*, vol. 8, no. 6, pp. 1524-1536, 1 Dec. 2022.

[30] J. Chen, et al., "A differential evolution-enhanced position-transitional approach to latent factor analysis," *IEEE Trans. on Emerging Topics in Computational Intelligence*, vol. 7, no. 2, pp. 389-401, April 2023.

[31] Z. Li, et al., "Diversified regularization enhanced training for effective manipulator calibration," *IEEE Trans. on Neural Networks and Learning Systems*, doi: 10.1109/TNNLS.2022.3153039.

[32] X. Luo, et al., "Adjusting learning depth in nonnegative latent factorization of tensors for accurately modeling temporal patterns in dynamic QoS data," *IEEE Trans. on Automation Science and Engineering*, vol. 18, no. 4, pp. 2142-2155, Oct. 2021.

[33] D. Wu, et al., "A double-space and double-norm ensembled latent factor model for highly accurate web service QoS prediction," *IEEE Trans. on Services Computing*, vol. 16, no. 2, pp. 802-814, 2023.

[34] H. Wu, et al., "A PID-incorporated latent factorization of tensors approach to dynamically weighted directed network analysis," *IEEE/CAA Journal of Automatica Sinica*, vol. 9, no. 3, pp. 533-546, March 2022.

[35] D. Wu, et al., "An L1-and-L2-norm-oriented latent factor model for recommender systems," *IEEE Trans. on Neural Networks and Learning Systems*, vol. 33, no. 10, pp. 5775-5788, 2022.

[36] D. Wu, et al., "A prediction-sampling-based multilayer-structured latent factor model for accurate representation to high-dimensional and sparse data," *IEEE Trans. on Neural Networks and Learning Systems*, 2022, doi: 10.1109/TNNLS.2022.3200009.

[37] X. Luo, et al., "A fast non-negative latent factor model based on generalized momentum method," *IEEE Trans. on Systems, Man, and Cybernetics: Systems*, vol. 51, no. 1, pp. 610-620, Jan. 2021.

[38] D. Wu, Y. He and X. Luo, "A graph-incorporated latent factor analysis model for high-dimensional and sparse data," *IEEE Trans. on Emerging Topics in Computing*, doi: 10.1109/TETC.2023.3292866.